

Paper 1 - Set 07 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

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Worked solutions

Question 1 IND-02 · Indices and surds · spec MM1.1

Find the value of $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$.

Answer: C

The negative index inverts the fraction: $\left(\frac{8}{27}\right)^{2/3}$. Cube root first: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

Full method - how and why

A fractional **negative** index of a fraction. Three separate instructions are packed into $-\frac{2}{3}$: invert, take the cube root, square.

1 The minus sign inverts the fraction: $\left(\frac{27}{8}\right)^{-2/3} = \left(\frac{8}{27}\right)^{2/3}$.

why: $a^{-n} = \frac{1}{a^n}$, and the reciprocal of a fraction is the fraction turned upside down. Doing this first keeps the numbers small.

2 Take the cube root of top and bottom: $\left(\frac{8}{27}\right)^{1/3} = \frac{2}{3}$.

why: $a^{m/n} = (\sqrt[n]{a})^m$; rooting before powering keeps you working with 2 and 3 rather than 64 and 729.

3 Square the result: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

why: The numerator 2 in the index is the power.

Where it goes wrong

- Treating the minus sign as making the answer negative gives $-\frac{9}{4}$ or $-\frac{4}{9}$; a negative index never makes a positive number negative.
- Forgetting to square gives $\frac{2}{3}$; forgetting to invert gives $\frac{9}{4}$. All three are offered as options.

Takeaway Read a fractional index as three instructions - sign (invert), denominator (root), numerator (power) - and do them in that order.

Question 2 STA-08 · Statistics · spec M6.3

A cumulative frequency graph for 80 observations passes through (20, 20), (30, 40) and (45, 60). Estimate the interquartile range.

Answer: D

Lower quartile at cumulative frequency 20: value 20. Upper quartile at 60: value 45. IQR = 25.

Full method - how and why

On a cumulative frequency graph, quartiles are read by going **in** from the cumulative frequency axis at $\frac{n}{4}$, $\frac{n}{2}$, $\frac{3n}{4}$.

1 With $n = 80$: the lower quartile is at cumulative frequency $\frac{80}{4} = 20$, the upper at $\frac{3 \times 80}{4} = 60$.

why: Cumulative frequency graphs use $\frac{n}{4}$ and $\frac{3n}{4}$ directly (no +1, unlike a raw list).

2 The point (20, 20) gives $Q_1 = 20$.

why: Read across at CF = 20 and down to the value axis.

3 The point (45, 60) gives $Q_3 = 45$.

why: Read across at CF = 60.

4 IQR = $Q_3 - Q_1 = 45 - 20 = 25$.

why: The interquartile range is a difference of **values**, not of frequencies.

Where it goes wrong

- Subtracting the cumulative frequencies, $60 - 20 = 40$ - an offered option.
- Using the median point (30, 40) as a quartile.

Takeaway Go in from the frequency axis at $\frac{n}{4}$ and $\frac{3n}{4}$, read off values, then subtract the **values**.

Question 3 FUN-14 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ has a maximum point at $(3, 8)$. The graph of $y = f(x - a) + b$ has its maximum at $(1, 2)$. Find $a + b$.

Answer: D

The maximum moves from $(3, 8)$ to $(3 + a, 8 + b) = (1, 2)$, so $a = -2$, $b = -6$ and $a + b = -8$.

Full method - how and why

Follow the turning point through the transformation and match coordinates - each coordinate gives one equation.

- 1 $y = f(x - a) + b$ translates the graph a right and b up, so the maximum moves from $(3, 8)$ to $(3 + a, 8 + b)$.

why: A translation moves every point, including the turning point, by the same vector.

- 2 Match the x -coordinates: $3 + a = 1$, so $a = -2$.

why: A negative a simply means the shift is to the left.

- 3 Match the y -coordinates: $8 + b = 2$, so $b = -6$.

why: Same for the vertical shift.

- 4 Add: $a + b = -8$.

why: The question asks for the sum, not for either value alone.

Where it goes wrong

- Using $3 - a = 1$ and getting $a = 2$, which flips the sign of the answer.
- Answering -2 or -6 (one of the two constants) instead of their sum.

Takeaway $f(x - a) + b$ shifts by $(+a, +b)$; set up one equation per coordinate.

Question 4 QUA-15 · Quadratic functions · spec MM1.3

The graph of $y = x^2 + bx + c$ crosses the x -axis at $(1, 0)$ and $(5, 0)$. What is the y -coordinate of the vertex?

Answer: C

$y = (x - 1)(x - 5)$, vertex at $x = 3$: $y = (2)(-2) = -4$.

Full method - how and why

Given the two roots, use the factorised form and the symmetry of the parabola - no expansion needed.

- 1 Write the factorised form: $y = (x - 1)(x - 5)$.

why: Roots at 1 and 5 with leading coefficient 1 (given, since the equation is $x^2 + bx + c$).

- 2 The vertex lies midway between the roots: $x = \frac{1+5}{2} = 3$.

why: A parabola is symmetric, so its turning point sits on the axis of symmetry, exactly halfway between the roots.

- 3 Evaluate there: $y = (3 - 1)(3 - 5) = 2 \times (-2) = -4$.

why: Substituting into the factorised form is far quicker than expanding to $x^2 - 6x + 5$.

Where it goes wrong

- Averaging the roots but then substituting into the wrong expression, e.g. getting -9 from $(3)^2 - 6(3) + 5$ miscomputed.
- Giving the x -coordinate 3 instead of the y -coordinate.

Takeaway Midpoint of the roots gives the axis of symmetry; substitute into the factorised form for the vertex height.

Question 5 PRO-07 · Probability · spec M7.6

Three fair coins are tossed. Find the probability of getting more heads than tails.

Answer: F

Need 2 or 3 heads: $\frac{3+1}{8} = \frac{1}{2}$ (by symmetry between heads and tails).

Full method - how and why

Either count the favourable outcomes out of 8, or exploit the symmetry between heads and tails.

1 With three coins there are $2^3 = 8$ equally likely outcomes.

why: Each coin independently doubles the number of outcomes.

2 'More heads than tails' means 2 or 3 heads: $\binom{3}{2} + \binom{3}{3} = 3 + 1 = 4$ outcomes.

why: With an odd number of coins there is no tie, so every outcome has a strict majority.

3 $P = \frac{4}{8} = \frac{1}{2}$.

why: Or instantly: heads and tails are symmetric and a tie is impossible, so each must have probability $\frac{1}{2}$.

Where it goes wrong

- Answering $\frac{3}{8}$ - the probability of exactly 2 heads, forgetting the all-heads case.
- Trying the symmetry argument with an **even** number of coins, where ties break it.

Takeaway An odd number of fair coins: $P(\text{more heads}) = \frac{1}{2}$ by symmetry, since ties cannot occur.

Question 6 TRG-03 · Trigonometric functions and equations · spec MM4.3

Find the exact value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

Answer: D

$$\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1.$$

Full method - how and why

Pure recall of exact values. (It is also $\sin(60^\circ + 30^\circ) = \sin 90^\circ$, but the addition formula is off-specification.)

1 Write down the exact values: $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$, $\sin 30^\circ = \frac{1}{2}$.

why: These come from the half-equilateral triangle with sides 1, $\sqrt{3}$, 2 - worth being able to redraw in seconds.

2 Substitute: $\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4}$.

why: $(\sqrt{3})^2 = 3$, so the first product is $\frac{3}{4}$.

3 Add: 1.

why: A clean answer - the sign that the exact values were right.

Where it goes wrong

- Using $\sin 60^\circ = \frac{1}{2}$ (that is $\sin 30^\circ$), giving $\frac{\sqrt{3}}{2}$.
- Writing $\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{\sqrt{3}}{4}$.

Takeaway sin and cos of 30° and 60° swap over; square a surd fraction by squaring top and bottom.

Question 7 IND-17 · Indices and surds · spec MM1.1

Solve $x^{\frac{3}{2}} = 8x^{-\frac{3}{2}}$ for $x > 0$.

Answer: E

Multiply by $x^{3/2}$: $x^3 = 8$, so $x = 2$.

Full method - how and why

An equation with the unknown in two fractional powers. Multiplying by the negative power clears it entirely.

- 1 Multiply both sides by $x^{3/2}$: $x^{3/2} \cdot x^{3/2} = 8$, i.e. $x^3 = 8$.

why: $x^{-3/2}$ becomes $x^0 = 1$ on the right, and the left gains $\frac{3}{2} + \frac{3}{2} = 3$. Multiplying by a positive quantity (we are told $x > 0$) cannot introduce false solutions.

- 2 Take the cube root: $x = 2$.

why: $x^3 = 8$ has one real root.

Where it goes wrong

- Cancelling the powers to get $x = 8$.
- Dividing by $x^{3/2}$ instead of multiplying, which leaves $x^3 = 8$ on the wrong side and invites sign errors in the index.

Takeaway To clear a negative fractional power, multiply through by that power - stated positivity guarantees the step is reversible.

Question 8 TRI-02 · Triangles, radians and sectors · spec MM4.1

A triangle has sides 4, 5 and 6. Find the cosine of its largest angle.

Answer: B

The largest angle is opposite the side 6: $\cos \theta = \frac{16 + 25 - 36}{2 \cdot 4 \cdot 5} = \frac{5}{40} = \frac{1}{8}$.

Full method - how and why

Three sides given, an angle wanted: the rearranged cosine rule. The word 'largest' tells you which angle.

- 1 Identify the largest angle: it is opposite the longest side, 6.

why: In any triangle the ordering of sides matches the ordering of opposite angles.

- 2 Use $\cos \theta = \frac{a^2 + b^2 - c^2}{2ab}$ with $a = 4$, $b = 5$, $c = 6$.

why: The two sides adjacent to the angle go on the bottom; the opposite side is the one subtracted on top.

- 3 Substitute: $\frac{16 + 25 - 36}{2(4)(5)} = \frac{5}{40} = \frac{1}{8}$.

why: Positive, so the largest angle is acute - this triangle has no obtuse angle at all.

Where it goes wrong

- Putting the longest side on the bottom, giving $\frac{9}{16}$ or similar.
- Assuming the largest angle must be obtuse and choosing $-\frac{1}{8}$.

Takeaway $\cos \theta = \frac{a^2 + b^2 - c^2}{2ab}$ with c opposite θ ; a positive result means an acute angle.

Question 9 PRO-06 · Probability · spec M7.7

In a class of 30, 18 study French, 12 study German and 5 study both. A student is chosen at random. Given that the student studies French, what is the probability that they also study German?

Answer: C

Conditional on French (18 students), 5 also study German: $\frac{5}{18}$.

Full method - how and why

The words 'given that' signal **conditional** probability: the sample space shrinks to the condition.

- 1 Restrict attention to the 18 French students - they are now the whole universe.

why: 'Given that the student studies French' means we only consider those students.

- 2 Of those 18, the number who also study German is 5.

why: The overlap is stated directly in the question.

- 3 $P(\text{German} | \text{French}) = \frac{5}{18}$
why: Equivalently $\frac{P(F \cap G)}{P(F)} = \frac{5/30}{18/30} = \frac{5}{18}$ - the 30s cancel.

Where it goes wrong

- Using $\frac{5}{30}$ - the unconditional probability of studying both, ignoring the condition.
- Using $\frac{5}{12}$, which conditions on **German** instead - the wrong way round.

Takeaway Conditioning replaces the denominator: divide by the size of the group you are told you are in.

Question 10 QUA-01 · Quadratic functions · spec MM1.3

The equation $x^2 + (k - 2)x + (k + 1) = 0$ has two distinct real roots. Find the complete set of possible values of k .

Answer: E

Discriminant $(k - 2)^2 - 4(k + 1) = k^2 - 8k = k(k - 8) > 0$, so $k < 0$ or $k > 8$.

Full method - how and why

'Two distinct real roots' is always a discriminant condition. The unknown is in the coefficients, so the discriminant becomes a quadratic in k and you finish with an inequality.

- 1 Identify $a = 1$, $b = k - 2$, $c = k + 1$ and write the discriminant: $(k - 2)^2 - 4(k + 1)$.

why: 'Two distinct real roots' $\Leftrightarrow b^2 - 4ac > 0$. Everything else in the question is decoration.

- 2 Expand and simplify: $k^2 - 4k + 4 - 4k - 4 = k^2 - 8k$.

why: The constants cancel, which is the sign you have expanded correctly.

- 3 Factorise and solve $k(k - 8) > 0$.

why: A quadratic inequality needs its roots and its shape: roots 0 and 8, opening upwards.

- 4 The upward parabola is positive outside its roots: $k < 0$ or $k > 8$.

why: Sketch it if in doubt - 'outside the roots' for > 0 , 'between the roots' for < 0 .

Where it goes wrong

- Expanding $(k - 2)^2$ as $k^2 - 4$, which gives $k^2 - 4k - 8$ and a different answer set.
- Reading the sign chart backwards and choosing $0 < k < 8$ - that is where the discriminant is **negative** and there are no real roots.

Takeaway Conditions on the number of roots are conditions on $b^2 - 4ac$; if the coefficients contain a parameter, you end up solving a quadratic inequality in that parameter.

Question 11 FUN-02 · Functions and transformations · spec MM8.2

$f(x) = x^2 - 4x$. The graph of $y = f(x + 2)$ has its minimum point at

Answer: A

$f(x) = (x - 2)^2 - 4$ has minimum at $(2, -4)$. Replacing x by $x + 2$ translates the graph 2 units to the left: $(0, -4)$.

Full method - how and why

Find the minimum of f first, then move it. Tracking the single turning point is quicker than re-completing the square.

- 1 Complete the square: $f(x) = x^2 - 4x = (x - 2)^2 - 4$, so f has its minimum at $(2, -4)$.

why: Half of -4 is -2 ; the -4 outside is the minimum value.

- 2 $y = f(x + 2)$ translates the graph 2 units to the left.

why: Replacing x by $x + a$ shifts left by a ; the graph now reaches each value 2 earlier.

- 3 Move the turning point: $(2, -4) \rightarrow (0, -4)$.

why: Only the x -coordinate changes; a horizontal translation never alters y .

Where it goes wrong

- Shifting right and answering $(4, -4)$.
- Changing the y -coordinate as well, giving $(0, -2)$ or similar.

Takeaway $f(x + a)$ moves the graph a to the left; horizontal shifts leave y -values untouched.

Question 12 STA-10 · Statistics · spec M6.3

For which of the following data sets is the mean greater than the median?

Answer: D

A large outlier pulls the mean above the median: mean $6 >$ median 3 . In B, D, E the mean equals the median; in C the mean $9.4 <$ median 11 .

Full method - how and why

The mean is pulled by extreme values; the median is not. Look for skew rather than computing everything.

- 1 Check 1, 2, 3, 4, 20: median $= 3$, mean $= \frac{30}{5} = 6$. Mean $>$ median. **Yes.**

why: The single large outlier 20 drags the mean far above the middle.

- 2 1, 2, 3, 4, 5 and 2, 4, 6, 8, 10 and 5, 5, 5, 5, 5 are symmetric, so mean $=$ median.

why: For symmetric data the two coincide - no need to compute.

- 3 1, 10, 11, 12, 13: median $= 11$, mean $= \frac{47}{5} = 9.4$, so mean $<$ median.

why: Here the outlier is small, so it drags the mean **down** - the mirror image.

- 4 So only 1, 2, 3, 4, 20 works.

why: One high outlier gives positive skew.

Where it goes wrong

- Assuming any outlier raises the mean - a low outlier lowers it, as in 1, 10, 11, 12, 13.
- Computing all five means when symmetry settles three of them instantly.

Takeaway Mean $>$ median signals a tail to the **right** (a high outlier); mean $<$ median signals a tail to the left.

Question 13 DIF-14 · Differentiation · spec MM6.2

Find $\frac{d^2y}{dx^2}$ when $y = \frac{1}{x^2} + 4\sqrt{x}$.

Answer: D

$$y' = -2x^{-3} + 2x^{-1/2}; y'' = 6x^{-4} - x^{-3/2}.$$

Full method - how and why

Two differentiations of powers, with negative and fractional indices. Rewrite first.

- 1 Rewrite: $y = x^{-2} + 4x^{1/2}$.

why: Roots and reciprocals must become powers before the rule applies.

- 2 First derivative: $y' = -2x^{-3} + 4 \times \frac{1}{2}x^{-1/2} = -2x^{-3} + 2x^{-1/2}$.

why: $-2 \times x^{-2-1}$ and $\frac{1}{2} \times 4 = 2$.

- 3 Second derivative: $y'' = -2(-3)x^{-4} + 2(-\frac{1}{2})x^{-3/2} = 6x^{-4} - x^{-3/2}$.

why: Differentiate again, watching the signs: -3 times -2 gives $+6$.

- 4 Write it as $\frac{6}{x^4} - x^{-3/2}$.

why: Matching the option format.

Where it goes wrong

- Losing a sign and answering $\frac{6}{x^4} + x^{-3/2}$.
- Stopping after the first derivative.

Takeaway Differentiate twice, carefully: each negative index gains one more negative factor.

Question 14 SEQ-04 · Sequences and series · spec MM2.2

The n th term of an arithmetic sequence is $u_n = 7 - 4n$. Find the sum of the first 30 terms.

Answer: A

$$u_1 = 3, u_{30} = 7 - 120 = -113. S_{30} = \frac{30}{2}(3 - 113) = 15(-110) = -1650.$$

Full method - how and why

You are given the n th term rather than a and d . Generate the first and last terms and use the $\frac{n}{2}(a + l)$ form.

- 1 First term: $u_1 = 7 - 4 = 3$. Common difference: the coefficient of n is -4 , so $d = -4$.

why: In $u_n = An + B$ the gradient A is the common difference - the sequence is linear in n .

- 2 Last term: $u_{30} = 7 - 120 = -113$.

why: Substitute $n = 30$ directly.

- 3 Use $S_n = \frac{n}{2}(a + l) = 15(3 + (-113)) = 15 \times (-110)$.

why: This version is quickest when the first and last terms are easy to compute.

- 4 Evaluate: -1650 .

why: Negative, as expected - most terms are negative.

Where it goes wrong

- Taking $d = 7$ or $a = 7$ from the formula rather than computing u_1 .
- Sign slips when adding $3 + (-113)$.

Takeaway For $u_n = An + B$: $d = A$ and $u_1 = A + B$; then either sum formula works.

Question 15 GRA-08 · Graphs of functions · spec MM8.7

How many real solutions does the equation $x^3 = 3x + 1$ have?

Answer: D

Sketch $y = x^3 - 3x - 1$: turning points at $x = \pm 1$ with $y(-1) = 1 > 0$ and $y(1) = -3 < 0$. A cubic whose local maximum is positive and local minimum is negative crosses the axis three times.

Full method - how and why

Counting roots of a cubic. Find the turning values and see whether they straddle the axis.

- 1 Rearrange to $f(x) = x^3 - 3x - 1 = 0$.
why: Everything on one side, so the question becomes 'how many roots?'.
- 2 Differentiate: $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$, so the turning points are at $x = \pm 1$.
why: A cubic with two turning points can have one, two or three roots.
- 3 Evaluate: $f(-1) = -1 + 3 - 1 = 1 > 0$ (local maximum) and $f(1) = 1 - 3 - 1 = -3 < 0$ (local minimum).
why: The **values** at the turning points decide the count.
- 4 The local maximum is above the axis and the local minimum below, so the curve crosses three times.
why: It comes up from $-\infty$, crosses, comes down past the axis, crosses, then rises again.

Where it goes wrong

- Trying to factorise - this cubic has no rational roots, so factorising wastes time.
- Answering 1 because 'a cubic always has at least one root' without checking the turning values.

Takeaway Cubic root count: find the turning values; three roots iff the local max and min lie on opposite sides of the axis.

Question 16 SIM-13 · Simultaneous equations and inequalities · spec MM1.5

How many integers x satisfy $x^2 < 5x + 24$ and $x^2 > 3x$?

Answer: E

$x^2 - 5x - 24 < 0 \Rightarrow (x - 8)(x + 3) < 0 \Rightarrow -3 < x < 8$. $x(x - 3) > 0 \Rightarrow x < 0$ or $x > 3$. Integers: $-2, -1$ and $4, 5, 6, 7$: six.

Full method - how and why

Two quadratic inequalities, then an integer count. The second one factorises without a constant term, which changes its shape.

- 1 First: $x^2 - 5x - 24 < 0 \Rightarrow (x - 8)(x + 3) < 0 \Rightarrow -3 < x < 8$.
why: Between the roots for < 0 .
- 2 Second: $x^2 - 3x > 0 \Rightarrow x(x - 3) > 0 \Rightarrow x < 0$ or $x > 3$.
why: Roots 0 and 3; outside the roots for > 0 . Do not divide by x - that would lose the $x < 0$ branch.
- 3 Intersect: from $(-3, 8)$ keep $(-3, 0)$ and $(3, 8)$.
why: Overlap both solution sets on one number line.
- 4 Count integers: $-2, -1$ from the first piece and $4, 5, 6, 7$ from the second: six.
why: Endpoints $-3, 0, 3, 8$ are all excluded by strict inequalities.

Where it goes wrong

- Dividing $x^2 > 3x$ by x to get $x > 3$, losing all the negative solutions.
- Including 0 or 3 in the count.

Takeaway Never divide an inequality by the variable; factorise instead, then count integers explicitly.

Question 17 GRA-04 · Graphs of functions · spec MM8.7

The line $y = x + c$ and the curve $y = \frac{1}{x}$ have no points of intersection. Find the complete set of values of c .

Answer: F

$x^2 + cx - 1 = 0$ always has discriminant $c^2 + 4 > 0$: two intersections for every c . So there are no values of c with no intersection.

Full method - how and why

A 'find the values' question whose honest answer is 'none'. Substituting shows the discriminant can never be negative.

- 1 Set the expressions equal: $x + c = \frac{1}{x}$.
why: Intersections satisfy both equations.
- 2 Multiply by x (noting $x \neq 0$, since $\frac{1}{x}$ is undefined there): $x^2 + cx - 1 = 0$.
why: Clearing the fraction gives a quadratic whose roots are the intersection x -values.
- 3 Compute the discriminant: $c^2 - 4(1)(-1) = c^2 + 4$.
why: The -1 constant is what makes the discriminant unavoidably positive.
- 4 $c^2 + 4 > 0$ for every real c , so there are always two intersections: no value of c avoids them.
why: Also check the roots are non-zero: their product is $-1 \neq 0$, so neither is the excluded $x = 0$.

Where it goes wrong

- Assuming a line and a hyperbola must sometimes miss, and choosing an interval.
- Sign error making the discriminant $c^2 - 4$, which would give $-2 < c < 2$ - an offered option.

Takeaway Do the algebra even when the answer 'feels' like an interval; $b^2 - 4ac$ with $ac < 0$ is always positive.

Question 18 SEQ-20 · Sequences and series · spec MM2.2

The integers from 1 to 100 inclusive are written down. What is the sum of those that are multiples of 3 or 5 (or both)?

Answer: E

Multiples of 3: $3 + \dots + 99 = 3(1 + \dots + 33) = 3 \times 561 = 1683$. Multiples of 5: $5(1 + \dots + 20) = 1050$. Multiples of 15: $15(1 + \dots + 6) = 315$. Total $1683 + 1050 - 315 = 2418$.

Full method - how and why

A counting-and-summing problem needing inclusion-exclusion: multiples of 15 belong to both lists and would otherwise be counted twice.

- 1 Multiples of 3 up to 100: $3 + 6 + \dots + 99 = 3(1 + 2 + \dots + 33) = 3 \times \frac{33 \times 34}{2} = 3 \times 561 = 1683$.
why: Factor out the 3 so the bracket is a plain triangular number.
- 2 Multiples of 5: $5(1 + \dots + 20) = 5 \times 210 = 1050$.
why: $\lfloor 100/5 \rfloor = 20$ terms.
- 3 Multiples of both, i.e. of 15: $15(1 + \dots + 6) = 15 \times 21 = 315$.
why: 'Or' in mathematics includes 'both', and multiples of 15 have been counted in each of the first two sums.
- 4 Inclusion-exclusion: $1683 + 1050 - 315 = 2418$.
why: Add the two sets, subtract the overlap once.

Where it goes wrong

- Forgetting to subtract the multiples of 15, giving 2733 - which is offered.
- Using $\text{lcm}(3, 5) = 8$ or 30 instead of 15.

Takeaway For 'multiples of a or b ', subtract the multiples of $\text{lcm}(a, b)$ once.

Question 19 SIM-08 · Simultaneous equations and inequalities · spec MM1.5

Which of the following inequalities has exactly the same solution set as $\frac{1}{x} > 2$?

Answer: B

If $x > 0$: $1 > 2x$, so $0 < x < \frac{1}{2}$. If $x < 0$: $\frac{1}{x}$ is negative and cannot exceed 2. So $0 < x < \frac{1}{2}$. (Multiplying by x without considering its sign is the classic error.)

Full method - how and why

An inequality with the variable in the denominator. You must not multiply by x without knowing its sign - split into cases instead.

- 1 Case $x > 0$: multiplying by x keeps the direction, giving $1 > 2x$, i.e. $x < \frac{1}{2}$. Combined with $x > 0$: $0 < x < \frac{1}{2}$.
why: Multiplying an inequality by a positive quantity is safe; the case assumption must be carried into the answer.
- 2 Case $x < 0$: then $\frac{1}{x} < 0 < 2$, so the inequality can never hold.
why: Rather than multiplying by a negative (and flipping), just observe the sign - a negative number is never greater than 2.
- 3 Case $x = 0$: $\frac{1}{x}$ is undefined, so $x = 0$ is excluded.
why: Always check the value that breaks the expression.
- 4 Combine: the solution set is exactly $0 < x < \frac{1}{2}$.
why: Only the first case contributes.

Where it goes wrong

- Multiplying through by x blindly to get $x < \frac{1}{2}$, which wrongly includes every negative number.
- Answering $x > 2$ by 'inverting' both sides without justification.

Takeaway Never multiply an inequality by an expression of unknown sign: split into cases, or bring everything to one side and analyse the sign of the fraction.

Question 20 INT-05 · Integration · spec MM7.1

Find the area of the finite region enclosed by the curve $y = x^2$ and the line $y = 2x + 3$.

Answer: E

Intersections: $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$. Area
 $= \int_{-1}^3 (2x + 3 - x^2) dx = \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3 = (9 + 9 - 9) - \left(1 - 3 + \frac{1}{3} \right) = 9 + \frac{5}{3} = \frac{32}{3}$.

Full method - how and why

Area between a line and a curve: integrate (upper – lower) between the intersection points.

- 1 Find the intersections: $x^2 = 2x + 3 \Rightarrow x^2 - 2x - 3 = (x - 3)(x + 1) = 0$, so $x = -1$ and $x = 3$.
why: These are the limits of the enclosed region.
- 2 Decide which is on top: at $x = 0$ the line gives 3 and the curve gives 0, so the line is above.
why: Testing one interior point settles the order - guessing wrong gives the negative of the answer.
- 3 Integrate the difference: $\int_{-1}^3 (2x + 3 - x^2) dx = \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3$.
why: Subtracting first, then integrating once, is quicker and less error-prone than two separate integrals.
- 4 Evaluate: at 3, $9 + 9 - 9 = 9$; at -1 , $1 - 3 + \frac{1}{3} = -\frac{5}{3}$. Difference: $9 + \frac{5}{3} = \frac{32}{3}$.
why: $-\frac{(-1)^3}{3} = +\frac{1}{3}$ - the double negative is the classic slip.

Where it goes wrong

- Integrating (curve – line) and giving $-\frac{32}{3}$, or taking the magnitude of the wrong piece.
- Sign error at the lower limit, giving $\frac{16}{3}$ or $\frac{28}{3}$.

Takeaway Area between curves = $\int(\text{upper} - \text{lower})$ between intersections; check which is upper with one test value.